case, the decay curve can be resolved into the individual components. The long-lived nuclide (^{187}W, line A) can be observed to have linear decay at times long enough for the shorter-lived species to have died. The decay line related to this species can be extrapolated back to $t = 0$ and this line subtracted from the observed decay curve. The resulting curve should be the linear decay due to the shorter-lived species in the sample (line B). For more complex mixtures, this process may be repeated until the curve is completely resolved into linear components.

Sometimes it is possible in mixed decay to observe preferentially the decay of one species by proper choice of detection technique. For example, a proportional counter may be used at an operating voltage that allows detection of α-decay ($\psi_\alpha > 0$) but excludes detection of β-decay ($\psi_\beta = 0$). By contrast a typical Geiger counter can be used for β-decay but does not detect α-radiation since the α-particles do not penetrate the window of the Geiger tube. These problems are discussed more extensively in Ch. 9.

5.13. RADIOACTIVE DECAY UNITS

The SI unit for radioactivity is the *Becquerel (Bq)*, and the activity is given in reciprocal seconds, s^{-1}:

$$1 \text{ Becquerel } (\text{Bq}) = 1 \text{ } (\text{disintegration}) \text{ } s^{-1} \tag{5.68}$$

The *measured count rate* (R, in (5.64)) is given in *counts per second* (cps) or per minute (cpm); the abbreviations are in German Impulse pro minute (Ipm), and in French coups de minute (c/m).

An earlier unit, still in some use, is the *Curie unit* (abbreviated Ci) defined as:

$$1 \text{ Curie } (\text{Ci}) = 3.7 \times 10^{10} s^{-1} (\text{Bq}) \tag{5.69}$$

The Curie unit was originally defined as the number of decays per unit time and gram of ^{226}Ra, assuming its half-life to be 1580 y.

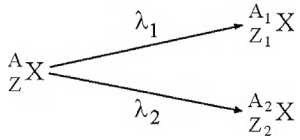
The *specific radioactivity* S is defined as the decay rate A per unit amount w of an element or compound,

$$S = A/w \tag{5.70}$$

The SI unit of specific radioactivity is Bq kg^{-1}. For practical purposes it is sometimes also defined in dpm g^{-1} or dpm $mole^{-1}$. *Activity concentration* (or "radioactive concentration") is given in Bq m^{-3} or Bq l^{-1}. With the half-life of 1599 ± 4 y the specific activity per gram of ^{226}Ra is 0.988 Ci or 3.65×10^{10} Bq or 2.19×10^{12} dpm. The specific activities of some of the longer-lived naturally occurring radioactive species are: ^{40}K, 31.0 kBq kg^{-1} of K; ^{232}Th, 4.06 MBq kg^{-1} of Th; ^{238}U, 12.3 MBq kg^{-1} of U.

5.14. BRANCHING DECAY

Several times in this chapter the possibility of competing modes of decay has been noted; see e.g. Figure 1.2. In such competition, termed *branching decay* (see e.g. Fig. 5.5(g)), the parent nuclide may decay to two or more different daughter nuclides: e.g.



(5.71)

where for each branching decay a *partial decay constant* can be determined. These constants are related to the total observed decay constant for the parent nuclide as

$$\lambda_{\text{tot}} = \lambda_1 + \lambda_2 + \ldots \tag{5.72}$$

Each mode of decay in branching may be treated separately; the decay in an individual branch has a half-life based on the partial decay constant. Since only the total decay constant (the rate with which the mother nuclide, ${}^{A}_{Z}\text{X}$ in (5.71), decays) is observable directly, partial decay constants are obtained by multiplying the observed total decay constant by the fraction of parent decay corresponding to that branch. ${}^{64}\text{Cu}$ decays 43% by electron capture, 38% by negatron emission, and 19% by positron emission. The observed total decay constant is equal to $0.0541\ \text{h}^{-1}$ based on the half-life of 12.8 h. The partial constants are:

$$\lambda_{\text{EC}} = 0.43 \times 0.0541 = 0.0233\ \text{h}^{-1} \tag{5.73a}$$

$$\lambda_{\beta^-} = 0.38 \times 0.0541 = 0.0206\ \text{h}^{-1} \tag{5.73b}$$

$$\lambda_{\beta^+} = 0.19 \times 0.0541 = 0.0103\ \text{h}^{-1} \tag{5.73c}$$

These partial decay constants correspond to partial half-lives of 29.7 h for electron capture decay, 33.6 h for β^- decay, and 67.5 h for positron decay.

5.15. SUCCESSIVE RADIOACTIVE DECAY

There are many instances where a parent decays to a daughter which itself decays to a third species (i.e. a "grand-daughter"). The chains of radioactive decay in the naturally occurring heavy elements include as many as 10−12 successive steps (Fig. 1.2).

$$\text{X}_1 \xrightarrow{\lambda_1} \text{X}_2 \xrightarrow{\lambda_2} \text{X}_3 \xrightarrow{\lambda_3} \text{X}_4 \ldots \tag{5.74}$$